NYC Taxi Dataset Processing: Benchmark Report

October 14, 2026

1 Overview

This report evaluates three methods for processing the NYC Taxi dataset (5.31 GB raw CSV, 55,423,856 rows, 8 columns): Standard In-Memory Pandas, Stream Processing via Chunking, and Parallelized Dask Compute.

2 Benchmark Results

Method	Time (s)	Peak Memory	Global Mean
Standard Pandas	597.72	2,664.25 MB	–
Stream Processing (Chunks)	121.33	112.53 MB	$11.35
Dask (Out-of-Core)	76.66	405.79 MB	$11.35

3 Key Findings

- **Standard Loading:** Successfully loaded into RAM but required nearly 10 minutes due to single-threaded parsing bottlenecks.
- **Stream Processing:** Maintained a very low memory footprint (112.53 MB) while running 5x faster than standard loading.
- **Dask Framework:** Delivered the fastest execution time (76.66 seconds) by leveraging multi-core parallelism and lazy evaluation.

4 When to Use Each Method

- **Standard Pandas:** Use only when the dataset fits comfortably into available RAM and you require simple, immediate, in-memory manipulations without complex infrastructure setups.
- **Stream Processing (Chunking):** Best for memory-constrained environments or when performing sequential, stateless row-by-row transformations and aggregations where maintaining a flat, low memory footprint is critical.
- **Dask:** Ideal for large datasets exceeding RAM limits when you need maximum performance, multi-core CPU scaling, and a familiar pandas-like API with lazy evaluation capabilities.